

Quality Assurance Testing Documentation (QATD)

BUILDORA AI: *AI-Powered Construction Workflow Automation*

UM HACKATHON 2026 - Domain 1: AI Agentic Workflow Automation

TEAM NAME

Dang Wangi

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Document Control

Field	Detail
System Under Test (SUT)	Buildora AI - AI-Powered Construction Workflow Automation
Team Repo URL	Github Link

Objective: The primary objective is to ensure that Buildora AI can reliably automate construction workflows, including document parsing, milestone monitoring, CIDB compliance checking, and report generation. The testing will focus on the accuracy of the five-agent orchestration system, state persistence in Firebase, and the reliability of automated alerts and reports under various conditions.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

This section aligns the testing back to specific user requirements from PRD v1.0 via a Requirement Traceability Matrix.

1.1 In-Scope Core Features

- **Agent A (Document Reader):** Parsing PDF/DOCX/Images and extracting 15+ structured fields.
- **Agent B (Schedule Monitor):** Detecting delays (>3 days) and cost overruns (>8%) from Firebase state.
- **Agent C (CIDB Compliance):** Validating project data against CIDB BISQ checklist and generating compliance scores.
- **Agent D (Alert & Notification):** Sending Telegram alerts within 10 seconds of escalation.
- **Agent E (Report Generator):** Generating branded PDF and XLSX reports in <10 seconds.
- **State Management:** Firebase Realtime Database persistence and inter-agent communication.
- **ChatBot Panel:** Conversational support interface allows users to submit natural language queries about their active project or about CIDB regulatory requirements.

1.2 Out-of-Scope

- Mobile application (Web responsive only).
- Real CIDB API integration (Mocked endpoints).
- Payment processing for subcontractors.
- Live IoT sensor data ingestion.

2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score	Mitigation Strategy	Testing Approach
GLM API Latency >6s	4	4	16 (Critical)	Implement pre-caching for demo; async streaming.	Load testing with concurrent API calls; monitor response times.
Extraction Accuracy <70%	3	5	15 (High)	Few-shot prompting; 3-stage fallback (pdfplumber/OCR).	Test with messy/scanned Malaysian construction docs.
Firebase State Handoff Bugs	3	5	15 (High)	Mandatory Day 2 gate; explicit state validation.	Integration testing of the LangGraph orchestrator.
CIDB Knowledge Base Gaps	2	4	8 (Medium)	Professional review; Day 9 Go/No-Go gate.	Manual verification of Agent C output against BISQ.

Risk Level Reference: * 1-5: Low (Monitor) * 6-10: Medium (Mitigate + Testing) * 11-15: High (Mitigate + Thorough Testing) * 16-25: Critical (Highest Priority; Extensive Testing)

3. Test Environment & Execution Strategy

- **Unit Test:**
 - **Scope:** Core agent functions (e.g., Agent B delay logic, Agent A field mapping).
 - **Execution:** pytest framework; runs on every push to GitHub.
 - **Isolation:** External dependencies (Firebase, GLM API) are mocked using `unittest.mock`.
- **Integration Test:**
 - **Scope:** LangGraph orchestrator flow (Agent A -> B -> C -> D -> E).
 - **Workflow:** Real Firebase calls in a staging environment; verified state transitions.
 - **Pass Condition:** Full pipeline completion from upload to report generation.
- **Test Environment (CI/CD):**
 - **Local:** Docker + Docker Compose for consistent local testing.
 - **Staging/CI:** GitHub Actions triggers pytest on PRs to main.
- **Passing Rate Threshold:**
 - **Critical Tests (P0):** 100% Passing Rate.
 - **Overall Test Suite:** Minimum 85% Passing Rate.

4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	Requirements
Automatic Build	Zero Build Errors
Unit Tests	100% Passing Rate for P0 features
Code Quality	Zero Linting Errors (Flake8/Black)
Test Coverage	Minimum 80% Coverage

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements
Regression Test	Minimum 95% Passing Rate
AI Output Pass Rate	Minimum 80% accuracy on extraction
Critical Bugs	Zero P0/P1 Bugs
API Performance	Response Time < 800ms (excluding LLM calls)

5. Test Case Specifications (Drafts)

Test Case ID	Test Type	Description	Test Steps	Expected Result
TC-01	Happy Case (Full Flow)	Verify end-to-end pipeline from upload to report.	1. Upload contract PDF. 2. Wait for Agent A extraction. 3. Trigger Agent B monitor. 4. Generate Report.	PDF report generated with correct extracted data and branded layout.
TC-02	Negative Case	Handle malformed/empty PDF upload.	1. Upload a 0KB PDF file. 2. Observe system response.	System returns "Invalid File" error; no crash; state remains stable.
TC-03	NFR (Performance)	Telegram alert latency.	1. Manually inject a >3 day delay in Firebase. 2. Measure time to Telegram notification.	Telegram alert received in <10 seconds.
TC-04	NFR (Load)	Concurrent report generation.	1. Trigger 10 report generation requests simultaneously.	All 10 reports generated in <20 seconds total; no 500 errors.

6. AI Output & Boundary Testing

6.1 Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)
AI-01	Extract project value from: “Contract Sum: RM 5,000,000.00”	JSON field project_value = 5000000.00
AI-02	Translate “Project Delay” to Bahasa Malaysia.	“Kelewatan Projek”
AI-03	Check compliance for missing “Site Safety Plan”.	Gap list includes “Site Safety Plan”; Score < 100%.

6.2 Oversized/Larger Input Test

Fields	Details
Maximum Input Size	20 Pages / ~8,000 Tokens
Input used while testing	50-page construction manual
Expected Behavior	System chunks document by page; processes first 20 pages or returns “File too large” warning.

6.3 Adversarial/Edge Prompt Test

- **Input:** Uploading a document in a language other than English/BM (e.g., French).
- **Expected Behavior:** Agent A identifies unsupported language and flags for manual review instead of hallucinating fields.

6.4 Hallucination Handling

- **Strategy:** Agent A uses structured JSON output with a “confidence score” for each field. If confidence < 0.7, the field is flagged in the UI for PM verification. Agent E cross-references report summaries against the raw Firebase state.